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Department of Computer Science and Engineering
Software Engineering
UE24CS341A

Software Requirements Specification (SRS)

StreamHub – OTT Video Streaming Website

Version: 1.0

Author: TEAM -04

Date: 20-09-2026

Status: Draft for Review

Revision History

Version	Date	Author	Change Summary
1.0	20-09-2026	TEAM -04	Initial simplified SRS for StreamHub (website only, FastAPI + React)

Approvals

Role	Name	Signature / Date
Course Coordinator	Arpitha.K Assistant Professor,	
Project Guide	Arpitha.K Assistant Professor	

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional requirements, non-functional requirements, external interfaces, and verification criteria for StreamHub, a web-based OTT (Over-The-Top) video streaming platform. It is the authoritative reference for the development team, QA reviewers, and academic evaluators on what the system must do.

1.2 Scope

StreamHub lets registered users browse a categorised catalog of videos, search for titles, stream video on demand through a web browser, maintain a watchlist, rate titles, and subscribe to a paid plan. It also gives administrators a simple interface to upload and manage content and to view registered users. StreamHub is delivered as a single responsive website — there is no native mobile app or smart-TV app in this scope. Advanced OTT features (DRM-protected playback, multi-bitrate adaptive streaming pipelines, offline downloads, and live linear TV) are explicitly excluded to keep the system realistic for an academic build.

1.3 Audience

- Developers implementing the React frontend and FastAPI backend
- QA engineers designing and executing test cases
- Content administrators managing the catalog
- Academic evaluators / project guide assessing the deliverable

1.4 Definitions

OTT (Over-The-Top streaming), VOD (Video on Demand), JWT (JSON Web Token), API (Application Programming Interface), PII (Personally Identifiable Information), RTM (Requirements Traceability Matrix), SPA (Single Page Application).

2. Overall Description

2.1 Product Perspective

StreamHub is a single web application: a React single-page frontend that talks to one FastAPI backend service over a REST API, backed by a single relational database. Video files are served from cloud/static storage. The system integrates with one external payment gateway for subscription payment. This is a simple client–server (3-tier) system, not a microservice or multi-client platform.

2.2 Major Product Functions

- Account registration, login, and profile management
- Catalog browsing and search
- Video playback in the browser
- Watchlist and star ratings
- Subscription plan selection and payment
- Content upload and publishing (admin)

2.3 User Roles and Characteristics

- Viewer: a registered user who browses and streams content from the website.
- Content Administrator: uploads and manages catalog titles and views basic user/subscription data.

2.4 Operating Environment

Client: any modern desktop or mobile web browser (Chrome, Edge, Firefox, Safari), responsive down to tablet width. Server: FastAPI (Python) backend deployed as a single service, PostgreSQL database, static/cloud storage for video and image files. No native mobile or TV client is built or supported.

2.5 Constraints

- Delivered as a website only — no native mobile app, no smart-TV app.
- Card and payment details are never stored on StreamHub's own server; they are handled entirely by the third-party payment gateway's hosted checkout.
- Video is served as standard progressive HTTP video (single or a small number of fixed qualities); full adaptive-bitrate transcoding and DRM are out of scope for this version.
- Built and demonstrated within academic project resources (no dedicated CDN or multi-region infrastructure required).

3. External Interface Requirements

3.1 User Interfaces

A responsive React web UI with: home/browse page, search bar with results, title-detail page, an HTML5 video player page, watchlist page, subscription/checkout page, and an admin dashboard for content upload. Basic accessibility: readable contrast, alt text on images, captions field for video metadata.

3.2 Hardware Interfaces

None beyond a standard keyboard, mouse/touch, and browser-supported audio/video output. No set-top box, remote-control, or dedicated device integration.

3.3 Software Interfaces

- Payment Gateway API (e.g., Razorpay/Stripe test mode) — hosted checkout for subscription payment and payment-status callback.
- Email service (e.g., SMTP/transactional email API) for OTP verification and password reset links.

3.4 Communications

- HTTPS/TLS 1.2+ enforced on all traffic between the browser and the FastAPI backend.
- REST/JSON over HTTPS between the React frontend and the FastAPI backend.

4. System Features — Functional Requirements

Each functional requirement is uniquely identified as OTT-F-###, grouped by module, with a priority and reference test case ID (see the Requirements Traceability Matrix in Section 8).

4.1 Account, Authentication & Profile

Req ID	Requirement (shall...)	Priority	Test Case Ref
OTT-F-001	The system shall allow a new user to register with an email address and password, verified via a one-time email OTP.	High	TC-Auth-01
OTT-F-002	The system shall authenticate a returning user by email and password and issue a signed JWT session token.	High	TC-Auth-02
OTT-F-003	The system shall allow a user to reset a forgotten password via a time-limited email reset link.	Medium	TC-Auth-03
OTT-F-004	The system shall allow a logged-in user to view and edit basic profile details (name, avatar, email).	Low	TC-Auth-04

4.2 Catalog, Browse & Search

Req ID	Requirement (shall...)	Priority	Test Case Ref
OTT-F-010	The system shall display the catalog as browsable rows grouped by genre (e.g., Action, Drama, Comedy).	High	TC-Browse-01
OTT-F-011	The system shall let a user search titles by name or genre and display matching results.	High	TC-Search-01
OTT-F-012	The system shall show a title-detail page with poster, synopsis, cast, genre, and average rating.	High	TC-Browse-02

4.3 Playback

Req ID	Requirement (shall...)	Priority	Test Case Ref
OTT-F-020	The system shall stream the selected video in an in-browser HTML5 video player.	High	TC-Play-01
OTT-F-021	The system shall resume playback from the last watched position for that user and title.	Medium	TC-Play-02
OTT-F-022	The system shall record each viewing session in the user's watch history.	Medium	TC-Play-03

4.4 Personalisation

Req ID	Requirement (shall...)	Priority	Test Case Ref
OTT-F-030	The system shall allow a user to add or remove a title from their personal Watchlist.	Medium	TC-Personal-01
OTT-F-031	The system shall allow a user to rate a title on a 1–5 star scale.	Low	TC-Personal-02

4.5 Subscription & Payment

Req ID	Requirement (shall...)	Priority	Test Case Ref
OTT-F-040	The system shall offer at least two subscription plans (e.g., Basic, Premium) with a listed price.	High	TC-Sub-01
OTT-F-041	The system shall process a subscription payment through a single integrated payment gateway's hosted checkout and mark the subscription active on a successful payment callback.	High	TC-Sub-02
OTT-F-042	The system shall display the user's current plan and renewal/expiry date on their account page.	Medium	TC-Sub-03

4.6 Content Administration

Req ID	Requirement (shall...)	Priority	Test Case Ref
OTT-F-050	The system shall allow an administrator to upload a title with metadata (name, synopsis, genre, poster image, video file).	High	TC-Admin-01
OTT-F-051	The system shall allow an administrator to publish or unpublish a title from the live catalog.	High	TC-Admin-02
OTT-F-052	The system shall allow an administrator to view a list of registered users and their subscription status.	Low	TC-Admin-03

5. Non-Functional Requirements

NFRs below are measurable and map to the Test Plan. IDs follow OTT-NF-###.

Req ID	Requirement	Category	Priority	Acceptance Criteria
OTT-NF-001	95% of API requests shall respond within 2 seconds under normal test load.	Performance	High	Measured in load test. Test: TC-Perf-01
OTT-NF-002	Video playback shall start within 5 seconds on a standard broadband connection.	Performance	Medium	Manual + scripted timing. Test: TC-Perf-02
OTT-NF-003	The website shall be usable on both desktop and tablet screen widths (responsive layout).	Usability	Medium	Manual check at 3 breakpoints. Test: TC-UI-01
OTT-NF-004	The system shall support at least 50 concurrent simulated users in the test environment without errors.	Scalability	Medium	Load test with 50 virtual users. Test: TC-Perf-03
OTT-NF-005	The backend shall be organised into clearly separated modules (auth, catalog, playback, subscription, admin) to support maintenance.	Maintainability	Low	Code review checklist. Test: TC-Maint-01
OTT-NF-006	The system shall be available during scheduled demo/test windows with no unplanned downtime.	Availability	Medium	Uptime log during test window. Test: TC-Avail-01

6. Security Objectives & Requirements

6.1 Security Objectives

- Protect user account credentials and personal information from unauthorised access.
- Ensure card/payment details are never handled or stored by StreamHub itself.
- Restrict administrative functions to authenticated administrators only.

6.2 Security Requirements

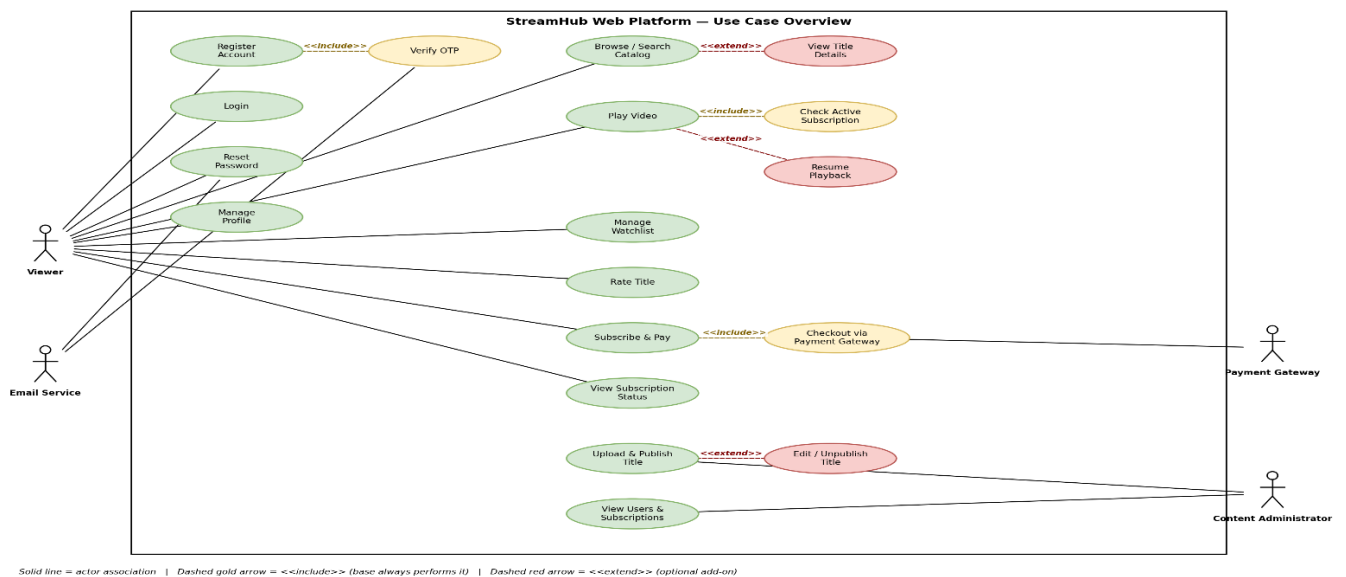
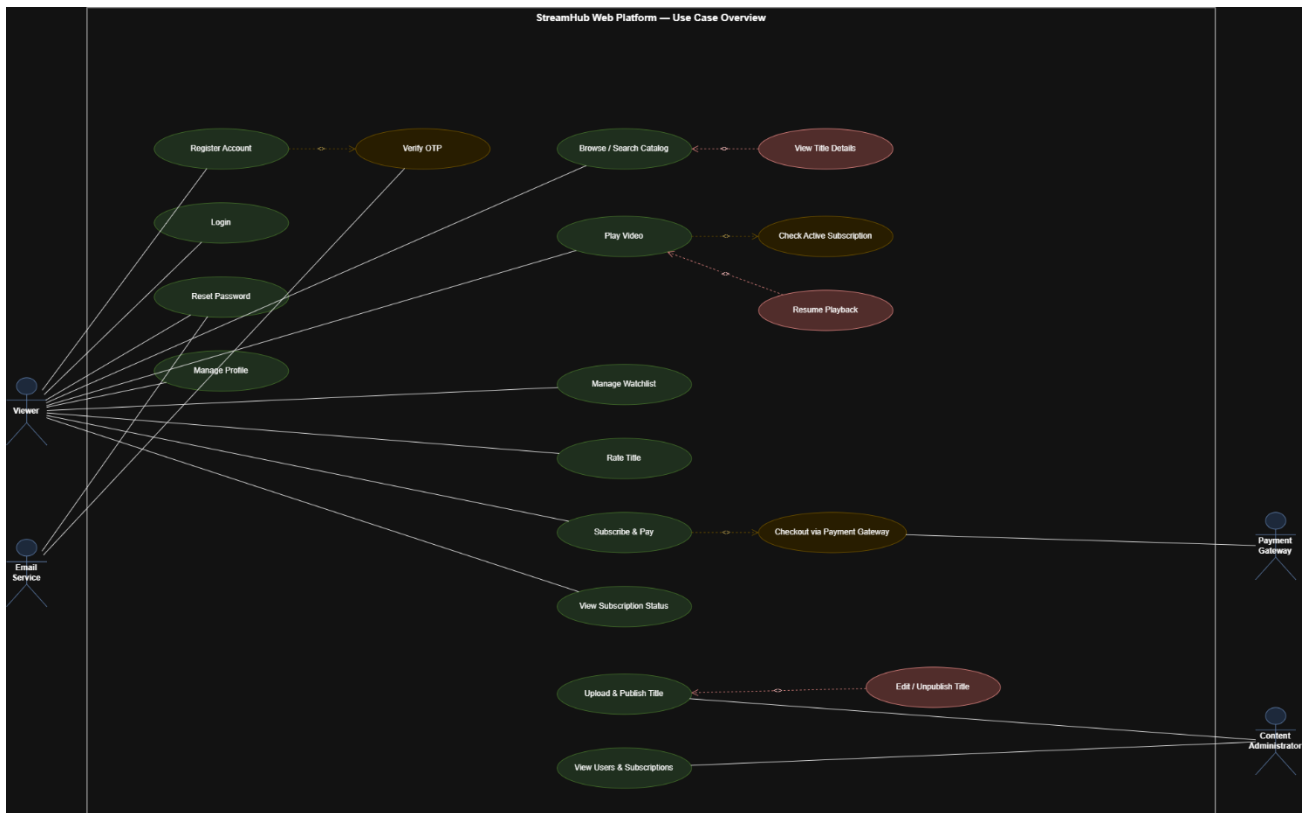
Req ID	Requirement (shall...)	Type	Priority
OTT-SR-001	All traffic between the browser and the backend shall use HTTPS/TLS 1.2+.	Security	High
OTT-SR-002	Passwords shall be stored using a salted hash (bcrypt); plaintext passwords shall never be stored or logged.	Security	High
OTT-SR-003	Session access shall use short-lived JWT tokens with an expiry, validated on every protected API call.	Security	High
OTT-SR-004	Card/payment data shall be entered only on the payment gateway's own hosted checkout page — it shall never pass through or be stored on StreamHub's server.	Security	High
OTT-SR-005	All API inputs shall be validated (using Pydantic models in FastAPI) to reject malformed or malicious input.	Security	Medium
OTT-SR-006	Admin-only API routes shall check the caller's role and reject requests from non-admin accounts.	Security	High

7. Quality Attributes & Acceptance Tests

Exit criteria for acceptance: all High-priority functional requirements implemented and verified, zero open critical defects, and the RTM showing all planned test cases passed.

Acceptance test suites: Authentication & Profile, Browse & Search, Playback, Subscription & Payment, Content Administration, Performance, and Security.

8. UML Diagrams



9. Requirements Traceability Matrix (RTM)

Req ID	Requirement (short)	Section	Test Case(s)	Status
OTT-F-001	Register account	4.1	TC-Auth-01	Not started
OTT-F-002	Login	4.1	TC-Auth-02	Not started
OTT-F-003	Password reset	4.1	TC-Auth-03	Not started
OTT-F-010	Browse catalog	4.2	TC-Browse-01	Not started
OTT-F-011	Search titles	4.2	TC-Search-01	Not started
OTT-F-020	Play video	4.3	TC-Play-01	Not started
OTT-F-021	Resume playback	4.3	TC-Play-02	Not started
OTT-F-030	Watchlist	4.4	TC-Personal-01	Not started
OTT-F-040	Subscription plans	4.5	TC-Sub-01	Not started
OTT-F-041	Payment gateway checkout	4.5	TC-Sub-02	Not started
OTT-F-050	Admin upload title	4.6	TC-Admin-01	Not started
OTT-F-051	Admin publish/unpublish	4.6	TC-Admin-02	Not started